

GET 208: STRENGTH OF MATERIALS

TORSION OF CIRCULAR MEMBERS

DEFINITIONS

1. Torque: is a moment that tends to twist a member about its longitudinal axis. Its effect is of primary concern in the design of axles or drive shafts used in vehicles and machines.

2. Torsion: If a bar is rigidly clamped at one end and twisted at the other by a torque applied in a plane perpendicular to its axis, the bar is said to be in torsion. The effects of a torsional load applied to a bar are:

a) To impart an angular displacement of one end cross-section with respect to the other end.

b) To set up shearing stresses on any cross-section of the bar perpendicular to its axis.

3. Shafts: These are usually cylindrical in section, solid or hollow. They are made up of mild steel, alloy steel and copper alloys. Shafts may be subjected to the following loads (a) torsional load (b) axial load (c) bending load (d) combination of above loads. The shafts are designed on the basis of strength and rigidity.

SHEAR STRESS DUE TO TORSION

To transmit energy by rotation it is necessary to apply a turning force. In case of a shaft, if the force is applied tangentially and in the plane of transverse cross-section, the torque or twisting moment may be calculated by multiplying the force with the radius of the shaft. If the shaft is subjected to two opposite turning moments it is said to be in pure torsion and there will be tendency of shearing off at every cross-section which is perpendicular to longitudinal axis.

When an external torque is applied to shaft it creates a corresponding internal torque within the shaft. An equation that relates this internal torque to the shear stress on the cross-section of a circular shaft is given as:

$$\tau_{max} = \frac{TR}{J}$$

where τ_{max} = Maximum shearing stress

T = Internal Torque

R = Radius of the shaft

J = Polar moment of Inertia

ANGLE OF TWIST

Occasionally, the design of a shaft depends on restricting the moment of rotation or twist that may occur when the shaft is subjected to a torque. In many problems such as torsion of shafts of milling and drilling machines, the angle of twist (θ) is required to be restricted. In such applications, the shafts are designed on the basis of limiting angle of twist and check for shearing stresses. If a shaft of length L is subjected to a constant twisting moment T along its length, then the angle through which one end of the bar will twist relative to the other is:

$$\theta = \frac{TL}{GJ}$$

where θ = Angle of twist
 T = Internal Torque
 L = Length of shaft
 G = Shear Modulus or modulus of rigidity
 J = Polar moment of Inertia

POLAR MOMENT OF INERTIA

For a hollow circular shaft of outer diameter D with a concentric hole of diameter d , the polar moment of inertia of the cross-sectional area, usually denoted by J , is given by:

$$J = \frac{\pi(D^4 - d^4)}{32}$$

The polar moment of inertia is obtained by setting $d = 0$.

EXERCISES

1) Calculate the minimum diameter of a solid shaft if it is to transmit a torque of 150,000 Nm without the minimum shear stress exceeding 45 N/mm².

(Ans: D = 257 mm)

2) In a torsion test, a piece of rod of 25 mm diameter was twisted 0.025 radian over a length of 200 mm, when a torque of 0.4 kNm was applied. Evaluate the modulus of rigidity for the material.

(Ans: G = 83.44 GN/m²)

3) A hollow shaft, 3 m long, is made up of steel and it is required to transmit a torque of 5000 Nm. If the total angle of twist is not to exceed 3° and the allowable shearing stress is 10,000 N/cm². Calculate the inner and outer diameters of the shaft. Take the modulus of rigidity, $G = 8.5 \times 10^6$ N/cm².

(Ans: D = 13.47 mm & d = 13.1 mm)

4) What must be the length of a 5 mm diameter aluminum wire, so that it can be twisted through one complete revolution without exceeding a shearing stress of 42 MN/m²? Take $G = 27$ GN/m².

(Ans: L = 10.1 m)